

Introduction

Usually, electromagnetic waves have their electric and magnetic field vectors oscillating in all directions. But using a polaroid, we can only let waves oscillating along a particular axis to pass.

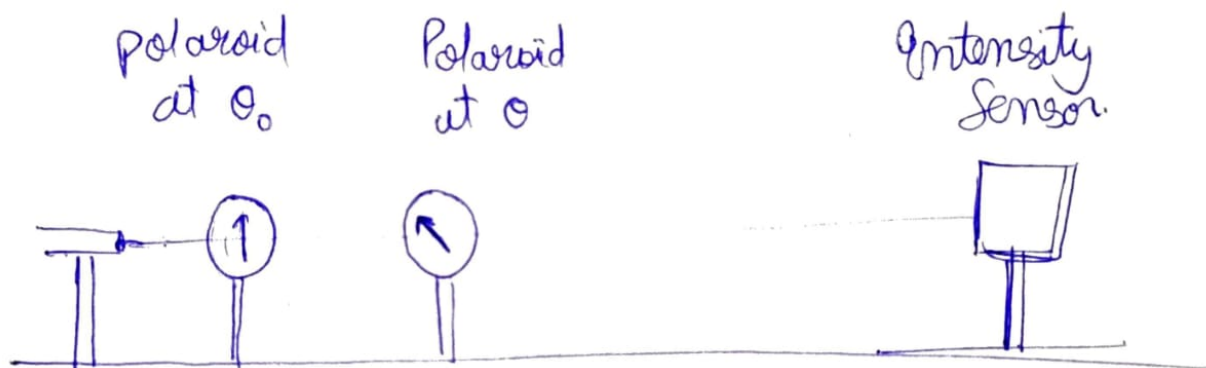
Further, given already polarized light, it is passed through another polaroid which is at an angle with respect to the light, Malus' Law tells us that the resulting intensity is directly proportional to the cosine squared of the angle. In this experiment, we verify Malus's Law

Experiment Setup and Procedure

Apparatus

Laser Source, Polaroid, intensity measuring device, stands for them all.

Experiment Setup



Procedure

The second polaroid is slowly turned and the angle for which the intensity measured is maximum is noted.

Then the polaroid is turned 10° each time and intensity is noted till the total angle turned is 90° and intensity is zero.

Then the graph of $\cos^2\theta$ and Intensity(I) is measured.

Formulae and Relations

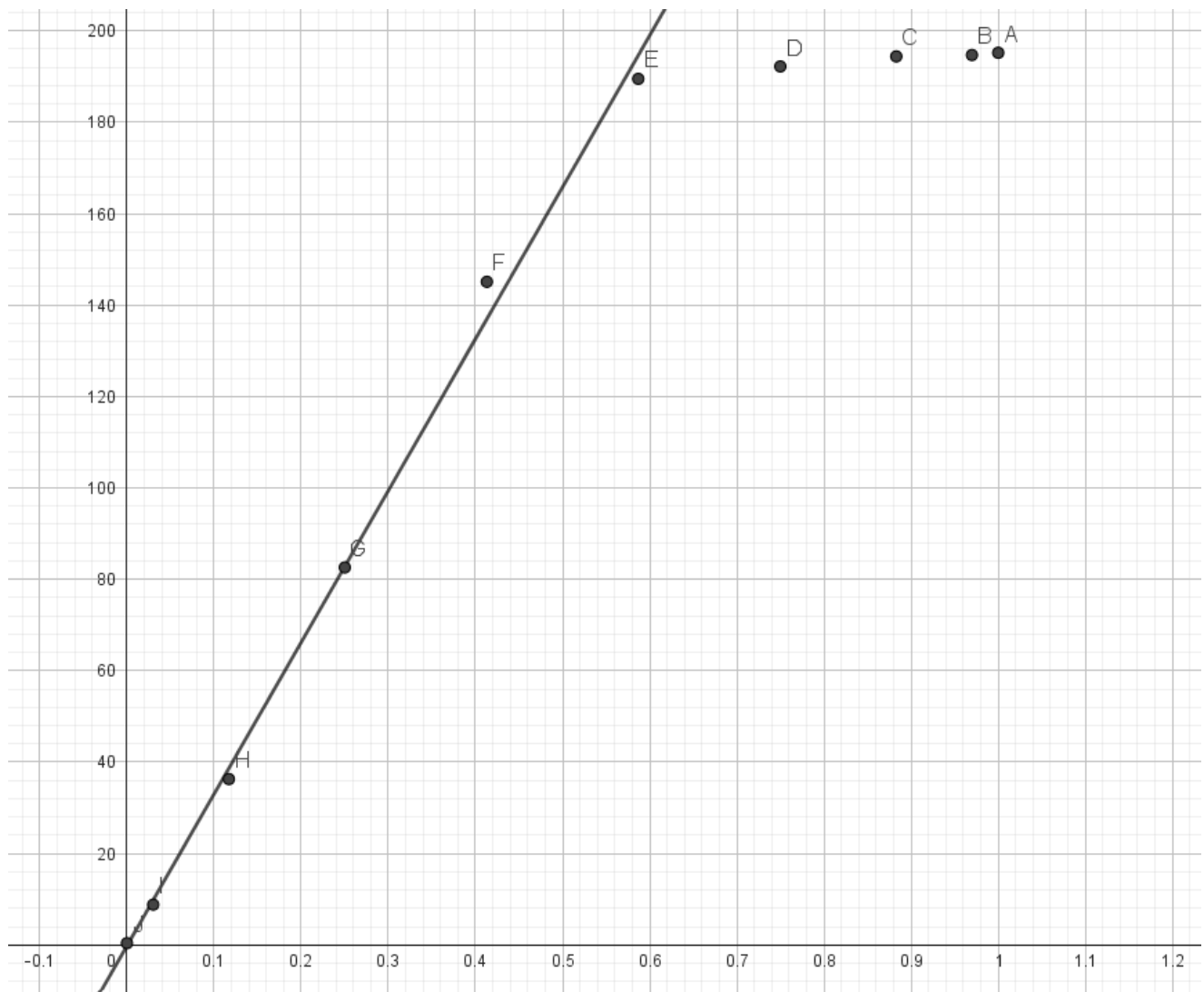
If θ is the difference between original the polarization of the light and the polaroid's angle, I_0 is the original intensity, and intensity of the light after passing through the polaroid is:

$$I = I_0 \cos^2 \theta$$

Data and Analysis

The Maximum value was found at $\theta_0 = 147^\circ$

θ	$\Delta\theta = \theta - \theta_0$	$\cos^2 \Delta\theta$	I
147	0	1	195.2
157	10	0.97	194.7
167	20	0.88	194.4
177	30	0.75	192.3
187	40	0.59	189.5
197	50	0.41	145.1
207	60	0.25	82.6
217	70	0.12	36.3
227	80	0.03	8.8
237	90	0	0.4



for larger values of $\Delta\theta$, i.e. when cos value is small, we get a nice linear graph. For larger values however, it seems the intensity become constant. This may be because the sensor reaches it's saturation point and cannot measure intensities about 200.